

VASHU KUMAR

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Education

National Institute of Technology Delhi

B.Tech in Computer Science Engineering

Graduating 2027

CGPA: 7.06

Dhruva Public School, Delhi

Senior Secondary Education

2021 – 2022

Percentage: 86%

Experience

Full-Stack Developer | Blinkit-style Delivery Ecosystem

Oct 2025 – Present

NIT Delhi (Under Director's Supervision)

- Engineered a 3-tier delivery system comprising dual **React Native** apps (Customer & Rider) and a **Next.js** admin dashboard for centralized inventory management.
- Architected a **Node.js & PostgreSQL** backend to synchronize real-time order lifecycles and ensure data consistency.
- Integrated **Firebase Cloud Messaging (FCM)** for automated push notifications, reducing communication latency.
- Developed secure RESTful APIs to handle multi-role authentication and dynamic inventory tracking for campus deployment.

Skills

- **Languages:** C/C++, Python, JavaScript/TypeScript, Kotlin, SQL, Bash
- **Backend & Cloud:** Node.js, Next.js, FastAPI, Docker, Azure, PostgreSQL, MongoDB, Redis, Celery
- **Architecture :** REST APIs, CI/CD, Event-Driven Systems, Webhooks
- **Libraries & Tools:** ReactJS, React Native, Jetpack Compose, Prisma, Git, GitHub, Postman
- **Specialized:** AST Parsing (Tree-sitter), LLM Integration, SAST

Projects

SDHD – AI-Powered GitHub PR Security Reviewer  | *Python, FastAPI, Celery, Redis, Gemini, Docker*

- Built a custom **GitHub App** that automatically analyzes pull requests on webhook events and posts inline security reviews.
- Implemented static analysis for Python and JS/TS using **AST** and **Tree-sitter** to detect dangerous functions and hardcoded secrets.
- Integrated **Gemini AI** to generate concise, context-aware remediation comments for detected vulnerabilities and logic hallucinations.
- Designed an asynchronous review pipeline with **Celery**, **Redis**, and HMAC verification, deployed via **Docker** on **Azure**.

YouFlex – Developer Social Platform  | *Next.js, JavaScript, PostgreSQL, Drizzle ORM*

- Engineered a full-stack social platform using **Next.js 16**, **NextAuth.js**, and **JWT**, backed by a serverless **Neon PostgreSQL** database managed through **Drizzle ORM**.
- Architected a paginated community feed using SQL limit/offset queries and **React's cache()** API to deduplicate database calls.
- Designed a relational schema with targeted **B-tree indexes** on high-frequency columns, reducing query overhead as data scales.
- Built a secure **Cloudinary** image upload pipeline and delivered a responsive UI with **Tailwind CSS v4** and dark mode.

Achievements & Extracurricular

- Solved 550+ DSA problems on **LeetCode**; active competitive programmer on **Codeforces**.
- Deputy General Secretary of Tech Club & Senior Member of T&P Cell (2024–June 2026).
- Participant in national-level hackathons including the Smart India Hackathon (SIH).